

# Solutions for Chapter 1

*Atmospheric and Oceanic Fluid Dynamics*

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1.1 For an infinitesimal volume, informally show that

$$\frac{D}{Dt}(\rho\varphi\Delta V) = \rho\Delta V \frac{D\varphi}{Dt}, \quad (S1.1)$$

where  $\varphi$  is some (differentiable) property of the fluid. Hence informally deduce that

$$\frac{D}{Dt} \int_V \rho\varphi dV = \int_V \rho \frac{D\varphi}{Dt} dV. \quad (S1.2)$$

The mass of an infinitesimal fluid element is  $\rho\Delta V$  and this is constant (by virtue of the definition of a fluid element). Thus, the factor can be taken out of the material derivative, and the result immediately follows.

The second result follows, informally, because the mass contained within the integral remains constant when taking the material derivative. Thus, (P1.2) is essentially the finite volume extension, or the integral, of (P1.1).

1.2 Show that the derivative of an integral is given by

$$\frac{d}{dt} \int_{x_1(t)}^{x_2(t)} \varphi(x, t) dx = \int_{x_1}^{x_2} \frac{\partial \varphi}{\partial t} dx + \frac{dx_2}{dt} \varphi(x_2, t) - \frac{dx_1}{dt} \varphi(x_1, t). \quad (S1.3)$$

By generalizing to three dimensions show that the material derivative of an integral of a fluid property is given by

$$\frac{D}{Dt} \int_V \varphi(\mathbf{x}, t) dV = \int_V \frac{\partial \varphi}{\partial t} dV + \int_S \varphi \mathbf{v} \cdot d\mathbf{S} = \int_V \left[ \frac{\partial \varphi}{\partial t} + \nabla \cdot (\mathbf{v}\varphi) \right] dV, \quad (S1.4)$$

where the surface integral ( $\int_S$ ) is over the surface bounding the volume  $V$ . Hence deduce that

$$\frac{D}{Dt} \int_V \rho\varphi dV = \int_V \rho \frac{D\varphi}{Dt} dV. \quad (S1.5)$$

Equation (P1.3) is known as Leibnitz's rule for differentiating an integral, or as differentiating under the integral sign. More-or-less long-winded proofs are given in elementary calculus books and on various web sites. In brief, if  $I(t) = \int_{x_1(t)}^{x_2(t)} \varphi(x, t) dx$  then by the chain rule

$$\begin{aligned} I'(t) &= \frac{\partial I}{\partial x_2} \frac{dx_2}{dt} + \frac{\partial I}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial}{\partial t} \int_{x_1}^{x_2} \varphi(x, t) dx \\ &= \varphi(x_2(t), t) \frac{dx_2}{dt} - \varphi(x_1(t), t) \frac{dx_1}{dt} + \int_{x_1}^{x_2} \frac{\partial}{\partial t} \varphi(x, t) dx. \end{aligned} \quad (S1.6)$$

The fundamental theorem of calculus is a special case of this formula, with  $x_1 = \text{constant}$ ,  $x_2(t) = t$  and  $\varphi(x, t) = \varphi(x)$ ; we have then

$$I'(t) = \varphi(t). \quad (\text{S1.7})$$

[In chapter 2 this expression is used in deriving (2.115).]

The extension to two and three dimensions is sometimes known to fluid dynamicists as the Reynolds transport theorem. It follows easily by proceeding in Cartesian geometry, and noting that  $dx/dt$  at the boundary is just the fluid velocity. Thus, terms arise of the form

$$\iint \frac{dx_2}{dt} \varphi(x_2(t), y, z, t) dy dz - \iint \frac{dx_1}{dt} \varphi(x_2(t), y, z, t) dy dz, \quad (\text{S1.8})$$

and similarly at the other boundaries. Then, much as in the elementary derivation of the mass continuity equation, we obtain the middle expression of (P1.4). The rightmost expression follows using the divergence theorem.

If we replace  $\varphi$  by  $\varphi\rho$ , then we can go from (P1.4) to (P1.5) by using the mass continuity equation in the form  $\partial\rho/\partial t + \nabla \cdot (\rho\mathbf{v}) = 0$ . Alternatively, and intuitively, (P1.5) follows directly as in problem 1.1 by noting that the mass element,  $\rho\Delta V$ , is constant.

- 1.3 *Why is there no diffusion term in the mass continuity equation? Suppose that a fluid contains a binary mixture of dry air and water vapour. Show that the change in mass of a parcel of air due to the diffusion of water vapour is exactly balanced by the diffusion of dry air in the opposite direction.*

There is no diffusion term because the net transport of mass is, by definition, taken into account by the fluid velocity. In a two-component fluid there may be diffusion of one component, such as water vapour or salt. But the centre of mass of any fluid parcel can only change, again by definition, if the fluid parcel moves. Thus, the diffusion of water vapour is exactly matched by diffusion of dry air in the opposite sense, so that diffusion as a whole does not change the centre of mass of the parcel.

- 1.4 *By invoking Galilean invariance we can often choose, without loss of generality, the basic state for problems in sound waves to be such that  $u_0 \equiv 0$ . The perturbation velocity is then certainly larger than the basic state velocity. How can we then justify ignoring the nonlinear term in the perturbation equation, as the term  $u'\partial u'/\partial x$  is certainly no smaller than the linear term  $u_0\partial u'/\partial x$ ?*

We would have to compare the nonlinear (or 'prime-prime') terms with the time derivative term, such as  $\partial\rho'/\partial t$ .

- 1.5 *What amplitude of sound wave is required for the nonlinear terms to become important? Is this achieved at a rock concert (120 dB), or near a jet aircraft that is taking off (160 dB)?*

In none of these cases is the amplitude large enough to make the sound waves nonlinear. Scaling the equations of motion suggests that sound waves may have a nonlinear affect when  $p' \sim \rho_0 c^2$ , or when  $p' \sim 10^6$  Pa. This corresponds to about 220db. If we take  $p' \sim 10^3$  Pa (or 10 mb, or about the pressure difference produced by a mild weather pattern), then this corresponds to about 150 dB.

- 1.6 Using the observed value of molecular diffusion of heat in water, estimate how long it would take for a temperature anomaly to mix from the top of the ocean to the bottom, assuming that molecular diffusion alone is responsible. Comment on whether you think the real ocean has reached equilibrium after the last ice age (which ended about 12 000 years ago).

The thermal molecular diffusivity of water is about  $1.3 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ . Using  $\partial \theta / \partial t = \kappa \nabla^2 \theta$ , the timescale for mixing,  $T_m$ , is given by  $T_m = H^2 / \kappa$ , where  $H$  is the distance in question, in this case 5 km (the average depth of the ocean). Thus,  $T_m \approx 5000^2 / 10^{-7} = 2.5 \times 10^{14} \text{ s} = 8 \times 10^6 \text{ years}$ . This is unrealistically long for at least two reasons: (i) Mixing in the ocean takes place using so-called eddy or turbulent diffusivities, which are about two orders of magnitude larger than the molecular value. (ii) Advection by the large scale will effectively mix the deep ocean on a still faster timescale. Still, the timescales are slow and the deep ocean may still remember the last ice age.

- 1.7 Consider the following flow:

$$u = \Gamma z, \quad v = V \sin[k(x - ct)],$$

where  $\Gamma$ ,  $V$ ,  $k$  and  $c$  are positive constants. (This is similar to the flow in the mid-latitude troposphere — an eastward flow increasing with height, with a transverse wave superimposed.) Suppose that  $\Gamma z > c$  for the region of interest. Consider particles located along the  $y = 0$  axis at  $t = 0$ , and compute their position at some later time  $t$ . Compare this with the streamfunction for the flow at the same time. (Hint: show that the meridional particle displacement is  $\eta = \psi / (u - c)$ , where  $\psi$  is the streamfunction and  $u$  and  $c$  are parameters.)

The kinematic relation for the meridional displacement is  $d\eta/dt = v$ , and linearizing this about the zonal flow gives

$$\frac{\partial \eta}{\partial t} + u \frac{\partial \eta}{\partial x} = v = \frac{\partial \psi}{\partial x}. \quad (\text{S1.9})$$

Looking for solutions of the form  $\eta = \eta_0 \exp[ik(x - ct)]$  gives

$$-ik(c - u)\eta_0 = ik\psi_0 \quad (\text{S1.10})$$

and so  $\eta = \psi / (u - c)$ . From the given flow field  $\psi = -(V/k) \cos[k(x - ct)]$  so that

$$\eta = -\frac{V}{k(u - c)} \cos[k(x - ct)]. \quad (\text{S1.11})$$

The meridional particle displacement is infinite at the critical level, where  $u = c$ .

- 1.8 ♦ Consider the two-dimensional flow:

$$u = A(y) \sin \omega t, \quad v = A(y) \cos \omega t.$$

The time-mean of this at a fixed point flow is zero. If  $A$  is independent of  $y$ , then fluid parcels move clockwise in a circle. What is its radius? If  $A$  does depend on  $y$ , find an approximate expression for the average drift of a particle,

$$\lim_{t \rightarrow \infty} \frac{\mathbf{x}(\mathbf{a}, t)}{t}$$

where  $\mathbf{a}$  is a particle label and variations in  $A$  are suitably ‘small’. Be precise about what small means.

If  $A$  is independent of  $y$  then we have  $dx/dt = A \sin \omega t$  and  $dy/dt = A \cos \omega t$ . These are the equations for clockwise motion in a circle, for note that they imply  $x^2 + y^2 = (A/\omega)^2$ , so the radius of the circle is  $A/\omega$ .

For small variations in  $A$  we write  $A(y) = A_0 + r(y - y_0)$ , where  $r = dA/dy|_{y=y_0}$ , and the  $v$  equation becomes

$$\frac{dy}{dt} = [A_0 + a(y - y_0) + O(r/\omega)] \cos \omega(t - t_0). \quad (S1.12)$$

The approximation is good if  $a|y - y_0| \ll A_0$ ; thus, in the region  $|y - y_0| \leq A_0/\omega$  it is a good approximation if  $r/\omega \ll 1$ . From (S1.12) we obtain

$$y - y_0 = \frac{A_0}{\omega} [\sin \omega(t - t_0) + O(r/\omega)] \quad (S1.13)$$

where  $t_0$  is the time at which  $y = y_0$  in the absence of shear.

We use the same approach for the equation for  $u$ , namely (and omitting the  $O()$  factors)

$$\frac{dx}{dt} = [A_0 + r(y - y_0)] \sin \omega(t - t_0), \quad (S1.14)$$

now using the (S1.13) for  $y$ . Thus, we solve

$$\frac{dx}{dt} = A_0 \left[ 1 + \frac{r}{\omega} \sin \omega(t - t_0) \right] \sin \omega(t - t_0). \quad (S1.15)$$

This may be integrated to give

$$x = x_0 - \frac{A_0}{\omega} \left\{ \cos \omega(t - t_0) - \frac{1}{2} r(t - t_0) + \frac{r}{4\omega} \sin 2\omega(t - t_0) \right\}. \quad (S1.16)$$

The second term in brackets gives what is known as ‘Stokes drift’, after George Stokes. Particles will drift in some direction, even when the time mean of the flow at a particular point is zero, as it is for the flow given in the question even when  $A = A(y)$ . The speed of the drift in the  $x$ -direction is

$$\overline{\left( \frac{x}{t} \right)} = \frac{1}{2} \frac{r A_0}{\omega}. \quad (S1.17)$$

and there is no drift in the  $y$ -direction. This problem illustrates the difference between the Eulerian and material perspectives.

- 1.9 (a) Suppose that a sealed, insulated container consists of two compartments, and that one of them is filled with an ideal gas and the other is a vacuum. The partition separating the compartments is removed. How does the temperature of the gas change? (Answer: it stays the same. Explain.) Obtain an expression for the final potential temperature, in terms of the initial temperature of the gas and the volumes of the two compartments.

Reconcile your answers with the first law of thermodynamics for an ideal gas, that

$$\delta Q = T d\eta = c_p \frac{d\theta}{\theta} = dI + \delta W = c_v dT + p d\alpha. \quad (S1.18)$$

- (b) A dry parcel that is ascending adiabatically through the atmosphere will generally cool as it moves to lower pressure and expands, and its potential temperature stays the same. How can this be consistent with your answer to part (a)?

- (a) The temperature stays the same because no work is done by the gas as it expands into the second chamber. (Experiments tend to show a slight cooling of the gas, because of non-ideality.) As regards the reconciliation, (S1.18) holds only for reversible changes, which the expansion is not.

The final potential temperature may be given by using the expression  $\theta = T(p_R/p)^{R/c_p}$ , where  $p_R$  is the reference pressure. We assume temperature stays the same, so that pressure falls in the ratio of the final volume to the initial volume. That is,  $p_f = p_i V_i/V_f$ , where the subscripts denote initial and final states. Thus,

$$\theta_f = T \left( \frac{p_R}{p_f} \right)^{R/c_p} = T \left( \frac{p_R V_f}{p_i V_i} \right)^{R/c_p} = \theta_i \left( \frac{V_f}{V_i} \right)^{R/c_p}. \quad (\text{S1.19})$$

- (b) During dry adiabatic ascent, the parcel is doing work against the ambient pressure as it expands. Thus, it cools and, if the process is reversible, the potential temperature (and the entropy) of the parcel stay the same.

1.10 Show that adiabatic flow in an ideal gas satisfies  $p\rho^{-\gamma} = \text{constant}$ , where  $\gamma = c_p/c_v$ .

One way to show this is to note that in such flows the potential temperature is constant. Thus

$$\theta = T \left( \frac{p_R}{p} \right)^{R/c_p} = \frac{p}{\rho R} \left( \frac{p_R}{p} \right)^{R/c_p} = \text{constant}. \quad (\text{S1.20})$$

Hence, noting that  $c_p - R = c_v$ ,  $p\rho^{-\gamma} = \text{constant}$ . Many thermodynamic textbooks give other ways of showing this.

1.11 (a) Show that for an ideal gas (1.116) is equivalent to (1.99). You may use the Maxwell relation  $(\partial\alpha/\partial\eta)_p = (\partial T/\partial p)_\eta$ .

(b) Show that for an ideal gas (1.127) is equivalent to (1.98).

(a) Two steps are necessary for this. First, use the mass continuity equation in the form  $D\rho/Dt = -\rho\nabla \cdot \mathbf{v}$  to eliminate  $D\rho/Dt$  in favour of  $\nabla \cdot \mathbf{v}$ . Second, to transform the right-hand side, after using the Maxwell equation given, you must evaluate  $(\partial T/\partial p)_\eta$ . For adiabatic flows,  $p^{r/c_p} T = \text{constant}$  so that  $\partial T/\partial p = -RT/c_p p$ , and the result follows.

(b) Here, the only thing that must be transformed is  $Dp/Dt$ , and for this we use the equation of state  $p = \rho RT$ , so  $Dp/Dt = \rho RDT/Dt + RTD\rho/Dt$ . Using  $\beta_T = 1/T$  for an ideal gas, the lhs of (1.127) becomes

$$\begin{aligned} \frac{DT}{Dt} - \frac{1}{c_p \rho} \frac{Dp}{Dt} &= \frac{DT}{Dt} - \frac{1}{c_p \rho} \left( \rho R \frac{DT}{Dt} + RT \frac{D\rho}{Dt} \right) \\ &= \frac{c_v}{c_p} \frac{DT}{Dt} - \frac{RT}{\rho c_p} \frac{D\rho}{Dt} = \frac{c_v}{c_p} \frac{DT}{Dt} + \frac{RT}{c_p} \nabla \cdot \mathbf{v} = \frac{c_v}{c_p} \frac{DT}{Dt} + \frac{p\alpha}{c_p} \nabla \cdot \mathbf{v} \end{aligned} \quad (\text{S1.21})$$

using  $c_p - c_v = R$  for an ideal gas.

1.12 Show that it follows directly from the equation of state,  $P = RT/\alpha$ , that the internal energy of an ideal gas is a function of temperature only.

From (1.87) and  $p = RT/\alpha$  we have

$$d\eta = \frac{1}{T} \left( \frac{\partial I}{\partial T} \right)_\alpha dT + \left[ \frac{1}{T} \left( \frac{\partial I}{\partial \alpha} \right)_T + \frac{R}{\alpha} \right] d\alpha. \quad (\text{S1.22})$$

But, mathematically,

$$d\eta = \left(\frac{\partial\eta}{\partial T}\right)_\alpha dT + \left(\frac{\partial\eta}{\partial\alpha}\right)_T d\alpha. \quad (S1.23)$$

Equating the coefficients of  $dT$  and  $d\alpha$  in these two expressions gives

$$\left(\frac{\partial\eta}{\partial T}\right)_\alpha = \frac{1}{T} \left(\frac{\partial I}{\partial T}\right)_\alpha \quad \text{and} \quad \left(\frac{\partial\eta}{\partial\alpha}\right)_T = \frac{1}{T} \left(\frac{\partial I}{\partial\alpha}\right)_T + \frac{R}{\alpha}. \quad (S1.24)$$

Noting that  $\partial^2\eta/(\partial\alpha\partial T) = \partial^2\eta/(\partial T\partial\alpha)$  we obtain

$$\frac{\partial^2 I}{\partial\alpha\partial T} = \frac{\partial^2 I}{\partial T\partial\alpha} - \frac{1}{T} \left(\frac{\partial I}{\partial\alpha}\right)_T. \quad (S1.25)$$

Thus,  $(\partial I/\partial\alpha)_T = 0$ . Because, in general, the internal energy may be considered either a function of temperature and density or temperature and pressure, this proves that for an ideal gas the internal energy is a function *only* of temperature.

- 1.13 *Show that it follows directly from the equation of state  $P = RT/\alpha$ , that for an ideal gas the heat capacity at constant volume,  $c_v$ , is, at most, a function of temperature.*

From the previous problem, the internal energy is a function only of temperature. From the next problem, the internal energy is given by  $I = \int c_v dT$ . Hence the heat capacity,  $c_v$ , is a function only of temperature.

- 1.14 *Show that for an ideal gas*

$$T d\eta = c_v dT + p d\alpha. \quad (S1.26)$$

*and that its internal energy is given by  $I = \int c_v dT$ .*

Let us regard  $\eta$  as a function of  $T$  and  $\alpha$ , where  $\alpha$  is the specific volume  $1/\rho$ . Then

$$\begin{aligned} T d\eta &= T \left(\frac{\partial\eta}{\partial T}\right)_\alpha dT + T \left(\frac{\partial\eta}{\partial\alpha}\right)_T d\alpha \\ &= c_v dT + T \left(\frac{\partial\eta}{\partial\alpha}\right)_T d\alpha \end{aligned} \quad (S1.27)$$

by definition of  $c_v$ . For an ideal gas the internal energy is a function of temperature alone (problem 1.12), so that using (1.70) the pressure of a fluid  $p = T(\partial\eta/\partial\alpha)_T = T(\partial\eta/\partial\alpha)_T$  and (S1.27) becomes

$$T d\eta = c_v dT + p d\alpha \quad (S1.28)$$

But, *in general*, the fundamental thermodynamic relation is

$$T d\eta = dI + p d\alpha. \quad (S1.29)$$

The terms on the right-hand side of (S1.28) are identifiable as the change in the internal energy and the work done on a fluid, and so  $dI = c_v dT$ . The heat capacity need not necessarily be constant, although for air it very nearly is, but it must be a function of temperature only.

- 1.15 (a) *Beginning with the expression for potential temperature for a simple ideal gas,  $\theta = T(p_R/p)^\kappa$ , where  $\kappa = R/c_p$ , show that*

$$d\theta = (\theta/T)(dT - (\alpha/c_p) dp), \quad (S1.30)$$

*and that the first law of thermodynamics may be written as*

$$\delta Q = T d\eta = c_p (T/\theta) d\theta. \quad (S1.31)$$

(b) Verify that  $d\eta = c_p(p_R, \theta)d\theta/\theta$ , for a fluid parcel of constant composition that obeys (1.147).

(a) The first part follows by straightforward substitution, using the ideal gas equation of state. The second part follows using (1.107) or (1.142), namely  $d\eta = c_p d\theta/\theta$  where  $c_p$  is constant.

(b) This follows with a little algebra. The detailed form of the equation of state barely enters. The main thing to note is the use of  $c_p(p_R, \theta)$  and not  $c_p(p, \theta)$ ; you won't be able to derive this without it.

1.16 Obtain an expression for the Gibbs function of a simple ideal gas, in terms of pressure and temperature.

The Gibbs function is given by  $G = I - T\eta + p\alpha$ . Using the fundamental equation of state, (1.113), namely  $\eta = c_v \ln(c_v T) - R \ln \rho + \text{constant}$ , we obtain

$$G = c_p T(1 - \ln T) + RT \ln p + AT \quad (\text{S1.32})$$

where  $A$  is a constant. (The Gibbs function is an extensive function, meaning that its value increases in proportion to the amount of substance present. Here we have calculated the Gibbs function for a unit mass. Note also we could combine the terms  $c_p T$  and  $AT$ .)

1.17 From (1.13) derive the conventional equation of state for an ideal gas, and obtain expressions for the heat capacities.

We are given that  $\eta = c_v \ln I + R \ln \alpha + \text{constant}$ , where  $R$  and  $c_v$  are constants. Then, using (1.70),

$$p = T \left( \frac{\partial \eta}{\partial \alpha} \right)_I = \frac{RT}{\alpha} = \rho RT. \quad (\text{S1.33})$$

Using (1.70b)

$$T^{-1} = \left( \frac{\partial \eta}{\partial I} \right)_\alpha = \frac{c_v}{I} \rightarrow I = c_v T. \quad (\text{S1.34})$$

The heat capacity at constant pressure,  $c_p$ , is

$$c_p \equiv T \left( \frac{\partial \eta}{\partial T} \right)_p = T \left( \frac{\partial}{\partial T} \right)_p [c_v \ln T + R \ln(RT/p)] = c_v + R. \quad (\text{S1.35})$$

and the heat capacity at constant volume, which we denote as  $C_v$ , is

$$C_v \equiv T \left( \frac{\partial \eta}{\partial T} \right)_\alpha = T \left( \frac{\partial}{\partial T} \right)_\alpha [c_v \ln T + R \ln \alpha] = c_v. \quad (\text{S1.36})$$

confirming our expectation and notation.

1.18 A parcel of water is added to the ocean surface that is denser than any water in the ocean. Suppose the parcel sinks adiabatically to the ocean bottom. Estimate the change in temperature that the parcel undergoes, being explicit about the assumptions you make.

Assume that the parcel conserves its potential temperature. Then let us use the linear equation of state (1.154c) to obtain

$$\Delta T = \frac{T_0 \alpha_0 \beta_T}{c_p} \Delta p \quad (\text{S1.37})$$

with  $\Delta\theta = 0$ . Using  $\Delta p = 5 \times 10^7$  (approx 5 km) and putting in numbers using table 1.2 gives

$$\Delta T = \frac{2839.7 \times 10^{-4} 1.7 \times 10^{-4}}{4 \times 10^3} 5 \times 10^7 \approx 0.6 \text{ K.} \quad (\text{S1.38})$$

We can also do the calculation from first principles, following the procedure on page 28, leading to (1.130) and (1.132). Then, for adiabatic flow, temperature changes according to  $\Delta T = -(\beta_T g T_0 / c_p) \Delta z$ , which leads to the same result as above.

- 1.19 Consider an ocean at rest with known vertical profiles of potential temperature and salinity,  $\theta(z)$  and  $S(z)$ . Suppose that we also know the equation of state in the form  $\rho = \rho(\theta, S, p)$ . Obtain an expression for the buoyancy frequency. Check your expression by substituting the equation of state for an ideal gas and recovering a known expression for the buoyancy frequency.

(This question requires some knowledge of chapter 2.) The buoyancy frequency is given by, in general

$$N^2 = -\frac{g}{\rho_\theta} \left( \frac{\partial \rho_\theta}{\partial z} \right), \quad (\text{S1.39})$$

where  $\rho_\theta$  is the locally referenced potential density of the environment. The potential density is given by an equation of state of the form  $\rho_\theta = \rho_\theta(\theta, S, p_R)$  where  $p_R$  is the reference pressure. Thus,

$$N^2 = -\frac{g}{\rho_\theta} \left( \frac{\partial \rho_\theta}{\partial \theta} \frac{\partial \theta}{\partial z} + \frac{\partial \rho_\theta}{\partial S} \frac{\partial S}{\partial z} \right). \quad (\text{S1.40})$$

For an ideal gas,  $S = 0$  and  $\rho_\theta = p_R / (R\theta)$ . We thus obtain  $N^2 = (g/\theta) \partial \theta / \partial z$ .

- 1.20 Consider a liquid, sitting in a container, with a free surface at the top (at  $z = H$ ). The liquid obeys the equation of state  $\rho = \rho_0[1 - \beta_T(T - T_0)]$ , and its internal energy,  $I$ , is given by  $I = c_p T$ . Suppose that the fluid is heated, and its temperature rises uniformly by  $\Delta T$ , and the free surface rises by a small amount  $\Delta H$ . Obtain an expression for the ratio of the change in internal energy to the change in gravitational potential energy (GPE) of the ocean, and show that it is related to the scale height (1.131). If global warming increases the ocean temperature by 4 K, what is the ratio of the change of GPE to the change of  $I$ ? Estimate also the average rise in sea level due to thermal expansion.

Suppose the container has unit area. Then the initial internal energy,  $I_1$  is given by  $I_1 = c_p \rho_1 H T_1$ . The final internal energy,  $I_2$  is given by  $I_2 = c_p \rho_2 (H + \Delta H) T_2 = c_p \rho_1 H T_2$ , using mass conservation. Thus

$$\Delta I = c_p \rho_1 H_1 (T_2 - T_1) = c_p \rho_1 H_1 \Delta T. \quad (\text{S1.41})$$

The initial GPE is given by

$$\text{GPE}_1 = \int_0^H \rho_1 g z \, dz = \frac{1}{2} \rho_1 g H^2 \quad (\text{S1.42})$$

and the final potential energy is given by

$$\text{GPE}_2 = \int_0^{H+\Delta H} \rho_2 g z \, dz = \frac{1}{2} \rho_2 g (H + \Delta H)^2 = \frac{1}{2} \rho_1 g H (H + \Delta H), \quad (\text{S1.43})$$



using the fact that, by mass conservation,  $\rho_1 H = \rho_2 (H + \Delta H)$ . Thus, the change in potential energy is given by

$$\Delta GPE = \frac{1}{2} \rho_1 g H_1 \Delta H = \frac{1}{2} \rho_1 g H_1 \beta_T H_1 \Delta T. \quad (S1.44)$$

Using the given equation of state, and mass conservation, we obtain

$$\Delta H = \frac{\rho_0 \beta_T \Delta T H_1}{\rho_2} \approx \beta \Delta T H_1. \quad (S1.45)$$

Using this in (S1.44) we obtain

$$\Delta GPE = \frac{1}{2} \rho_1 g H_1^2 \beta \Delta T \quad (S1.46)$$

Hence, using (S1.41) and (S1.46) we obtain  $\Delta GPE / \Delta I = g H_1 \beta_T / (2c_p)$ . This is equal to the ratio of the height of the fluid to the twice the temperature scale height of the fluid, as given by (1.131). For the ocean, the scale height is about 2000 km, so the ratio is very small. Thus, most of the heating goes into internal energy. In a Boussinesq fluid, or in general one which is almost incompressible, the internal energy is not available for conversion to kinetic energy.

Using (S1.44), the rise in sea-level for a 4 K temperature change is approximately  $\Delta H = \beta_T \Delta T H_1 \approx 1 \times 10^{-4} \times 4 \times 5000 = 2$  m, using a small value of  $\beta_T$  appropriate for the cool, abyssal ocean. This rise is much larger than the projected rise in sea level for the next century, even if the surface temperature does increase by 4 K, because the ocean will not warm all the way to the ocean floor on that time frame. However, it will eventually. . .

- 1.21 ♦ *Obtain an expression, in terms of temperature and pressure, for the potential temperature of a van der Waals gas, with equation of state  $(p + a/\alpha^2)(\alpha - b) = RT$ , where  $a$  and  $b$  are constants. Show that it reduces to the expression for an ideal gas in the limit  $a \rightarrow 0$ ,  $b \rightarrow 0$ .*

I haven't done this. However, the expression for the entropy of a van der Waals gas is derived in some thermodynamics books, for example Callen (1985) or Reif (1965). From that, one may obtain an expression for the potential temperature by solving the expression

$$\eta(T, p) = \eta(\theta, p_R). \quad (S1.47)$$

for  $\theta$ .